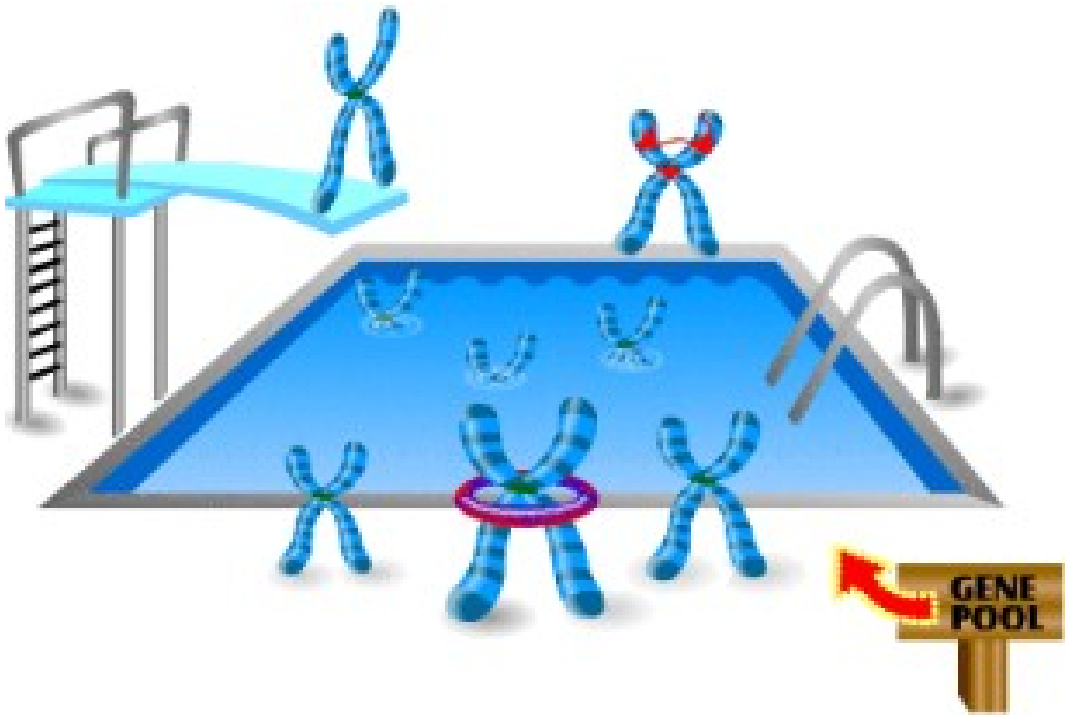
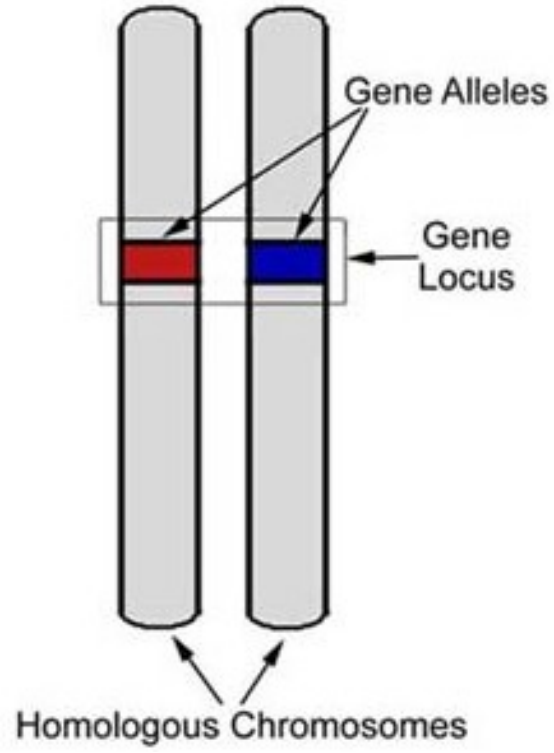
Change in Gene Pools over Time

Intro, Gene Mutations



Important Terms and Definitions: Alleles

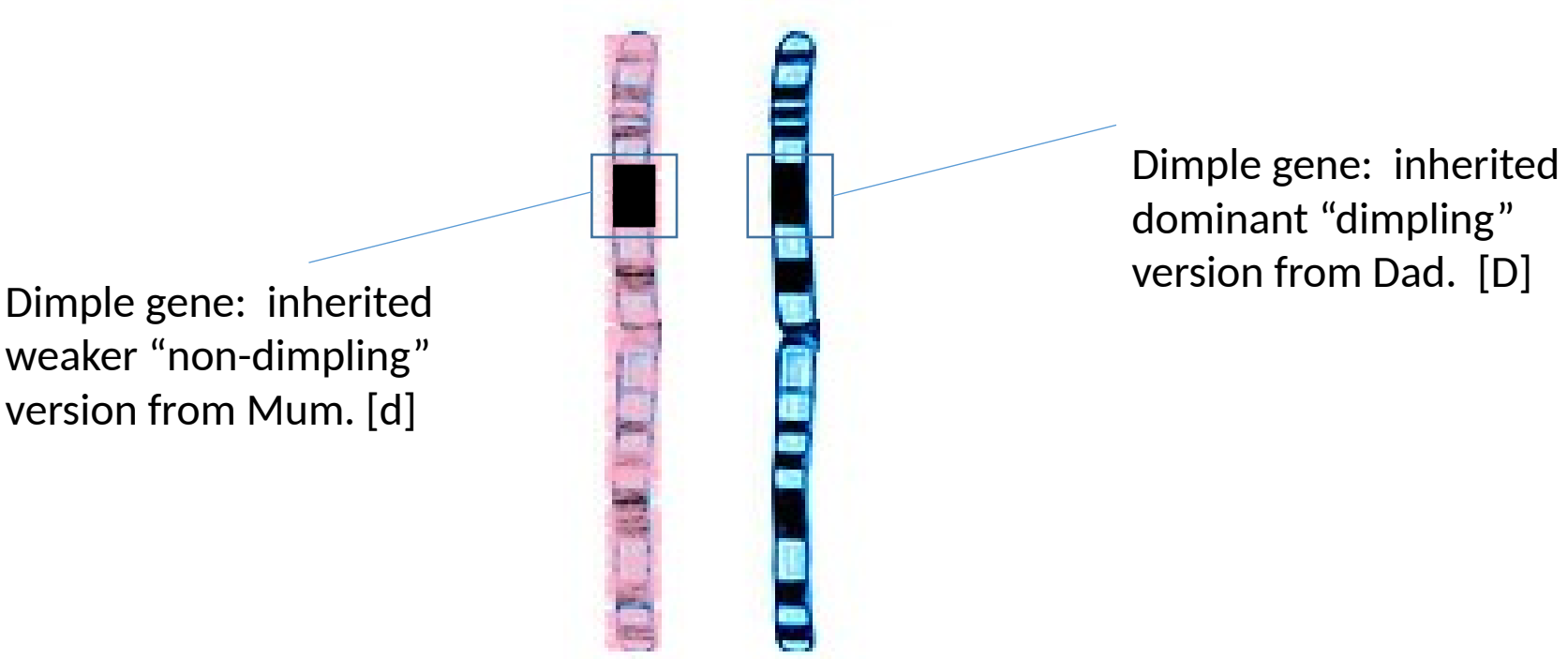
- Alleles are alternative forms of a gene.
- For any given gene, one allele (form) is inherited from the mother, and one from the father.



- The human genome contains many genes, each coding for different aspects of human structure and function.
- For each gene, two alleles are inherited.
- The traits an individual shows depends on the alleles inherited for each gene.

Learning Aim: Define key terms related to gene pools

Each half of a chromosome pair contains genes for the same things, but an individual has inherited different variants (alleles) from each parent:



Eg: This person has dimples because the dominant variant (allele) of the gene he got from his dad overcame the weaker variant from his mum. The “dimpling” gene is really a code giving instructions for protein production to the muscles in the cheek, which affects how those cells arrange themselves. One variant of the protein production results in no dimples. The other results in dimples.

Learning Aim: Define key terms related to gene pools

Important Terms and Definitions: Species

- Species: a group of individuals that share many characteristics and are able to breed under natural conditions to produce fertile offspring.

- Eg: Humans are members of the same species:

Homo sapiens

Chimpanzees are not the same genus or species as humans. They are of the genus and species *Pan troglodytes*

Neanderthals (extinct human ancestors) are from the same genus as modern humans, but a different species. They are *Homo neanderthalensis*

Learning Aim: Define key terms related to gene pools

Important Terms and Definitions: Population

- A population is a group of organisms of the same species, living together in a particular place at a particular time.
  - Eg: there is a population of human beings living in Perth right now, and another living in Sydney.
  - The population in Perth now is a different population than lived here 100 years ago.
- Can look at the genes, alleles and characteristics of a population as a whole, as well as individuals.



Learning Aim: Define key terms related to gene pools

Important Terms and Definitions: Gene Pool

- Gene Pool : The sum of all alleles in a given population

- ***Allele frequency:*** how often each allele for a gene occurs in a population.
  - Eg: the alleles for blond hair occur more frequently in Scandinavian populations than they do in Asian populations.
  - Eg: Allele for Cystic Fibrosis occurs on Chromosome Pair 7

    Cystic Fibrosis in Population N:

    Frequency of CF allele: 5%

    Frequency of normal allele: 95%

- Populations with different general characteristics are likely to have different allele frequencies, and therefore different gene pools.

Learning Aim: Define key terms related to gene pools

Change in gene pools over time

- Frequency and type of alleles in a population can change over time.

  - Mutation – can have no effect, produce an advantageous change or produce a disadvantageous change.
    - Gene Mutation
    - Chromosomal Mutation

  - Gene flow due to interbreeding between adjoining groups

  - Change to allele frequency via genetic drift
    - Random
    - Non-directional – frequency can drift back and forth

  - Natural selection -
    - alleles that provide an advantage to individuals in the environment increase in frequency.
    - alleles that provide no advantage decrease in frequency.
    - change to allele frequency is directional – increase in frequency of advantageous alleles.

Learning Aim: Outline ways in which frequency and type of alleles in a population can change over time

Mutations

- Changes to gene sequences or chromosomes
  - Occur during replication of DNA molecule before cell division.
  - Can be classified by:

WHAT CAUSES THEM TO OCCUR

- Spontaneous (occur randomly)
- Induced (are the result of a mutagen)

WHETHER THEY CAN BE INHERITED

- Somatic (non heritable)
- Germline (heritable)

THEIR EFFECT ON AMINO ACID SEQUENCE and PROTEIN PRODUCTION

- Missense, Nonsense, Neutral and Silent mutations have differing effects

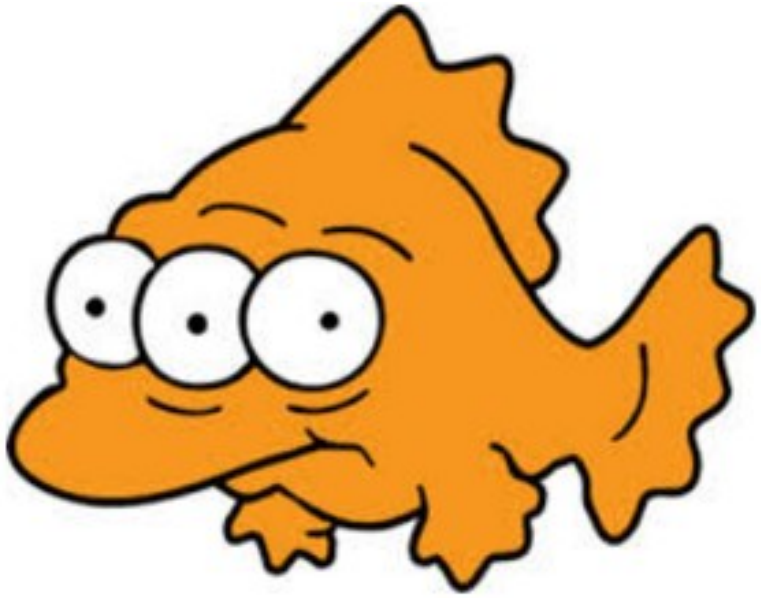
HOW MUCH DNA IS AFFECTED

- The base sequence for an individual gene (point mutation)
- Larger sections of DNA (chromosomal mutation)

Learning Aim: Define and classify mutations

Mutagens

- Some mutations arise by chance. These are *spontaneous mutations*.
- Others are caused by a *mutagen*. These are *induced mutations*.
- Mutagen: a substance or agent that affects the rate of mutation of DNA.
  - Ionising radiation:
    - UV light
    - X-rays
    - Nuclear radiation and more…
  - Some chemicals:
    - Mustard gas
    - Formaldehyde
    - Some antibiotics and other drugs
    - And more…

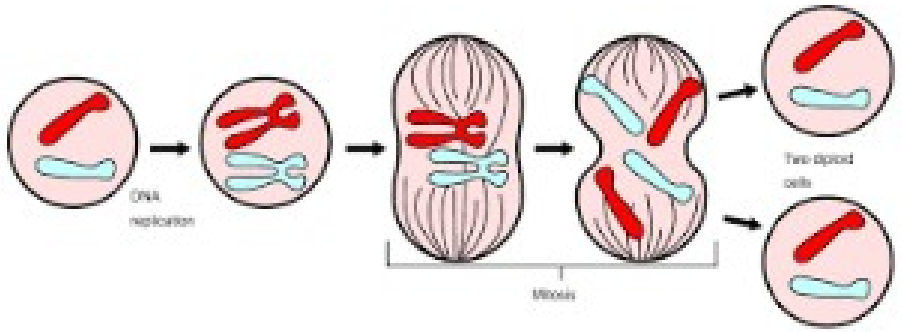


Learning Aim: Define mutagens and give examples

Somatic mutations

- Occur in non-reproductive cells, during mitosis – therefore affects only that individual, not offspring or population.
- May:
  - Have no effect on cell-line function
  - Cause cell death
  - Cause uncontrolled growth and structure of the cell-line tissue: cancer.

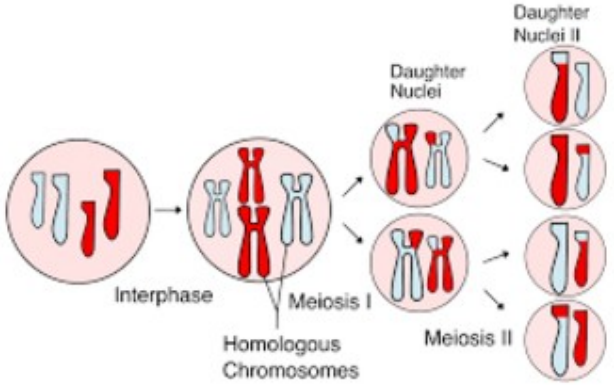
- Mutation is not passed down to further generations of organism: the sperm or egg cells aren't affected.



Learning Aim: Outline somatic mutations and their effects

Germline Mutations

- Germline cells = reproductive cells/gametes eg sperm and egg
- Germline mutations occur in gamete production, during meiosis
- Changes may:
  - Be passed on to offspring, subsequent generations and population over time.
  - Result in a non-viable gamete
  - Have no obvious effect
  - Have a negative effect: abnormalities, inherited disease
  - Have a positive effect: advantage for the individual and therefore for offspring and the population over time.



Learning Aim: Define germline mutations and describe the possible effects.

Gene (point) mutations

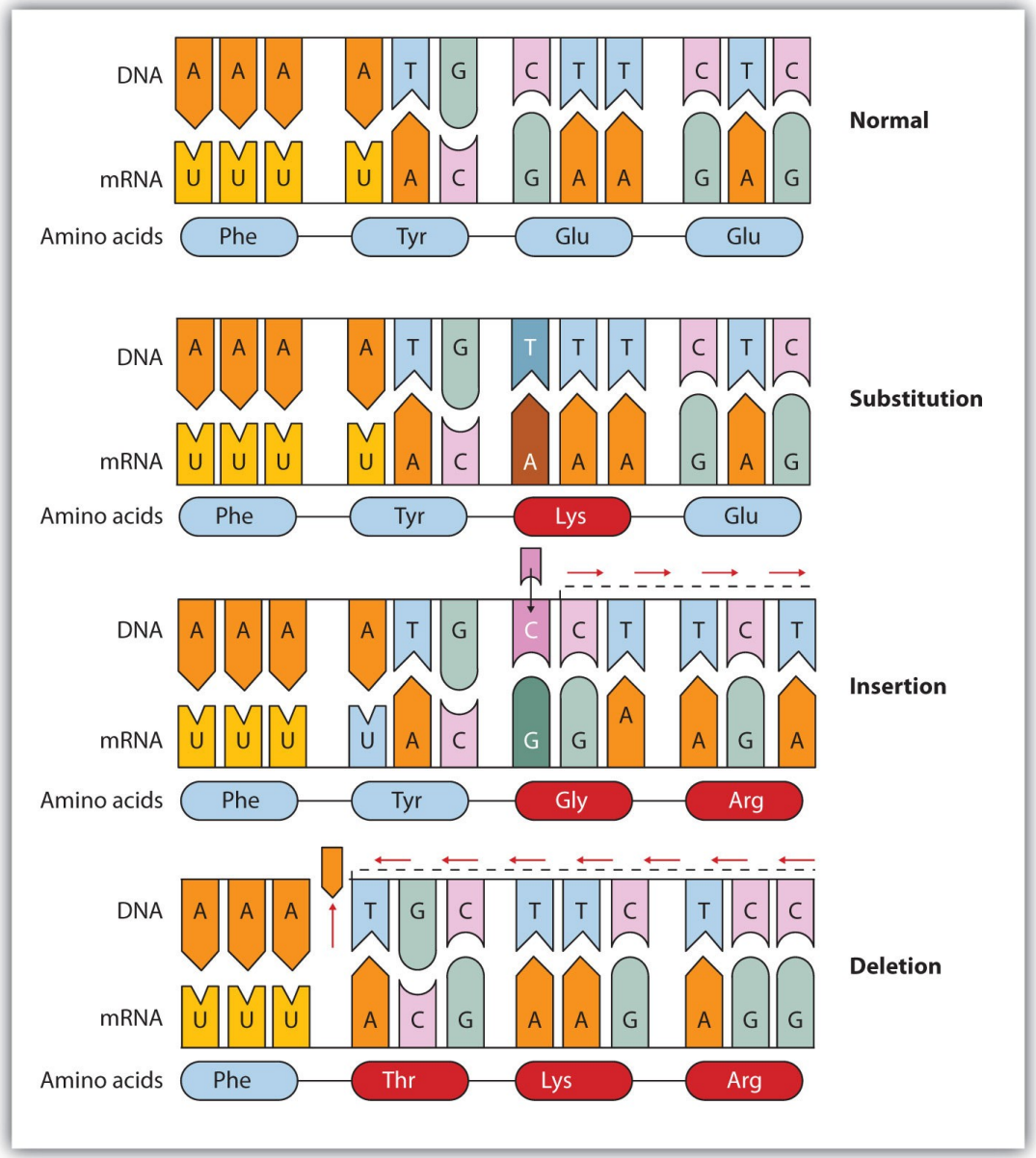
- Affect individual genes (not large chromosome chunks)
- Change the base sequence in the genetic code.
- Can alter structure and function of the protein made using the code.
- Change in the base sequence can be:

  Insertion: an additional nucleotide is added to the base sequence.

  Deletion: a nucleotide is removed from the base sequence.

  Substitution: a nucleotide is replaced with a different nucleotide.

  Insertion and deletion will result in a *frame shift* – they affect every subsequent triplet and therefore the amino acids coded for.



Learning Aim: List and describe the three types of point mutation.

Gene (point) Mutation Effects

- Change the base sequence in the genetic code.
- Can alter structure and function of the protein made using the code.
- May have no effect, may cause problems, or may be an advantage.
  - *Missense mutations* cause a change in the amino acid and therefore the protein produced
  - *Nonsense mutations* change the base sequence to stop amino acid addition causing the protein to lose structure and function
  - *Neutral mutations* cause change in the amino acid but it does not affect protein structure
  - *Silent mutations* cause change in base sequence but no change to amino acid or protein – most amino acids are coded for by multiple triplet combos.

Learning Aim: Describe the effects of point mutations on amino acid production and therefore protein structure.

Inherited disease from gene mutation

- Base sequence mutation in the germline cells can cause a protein to become faulty or absent, which can result in a disease inherited with each new generation.

- These diseases are often *recessive* – both inherited alleles must be faulty for the person to develop the disease.

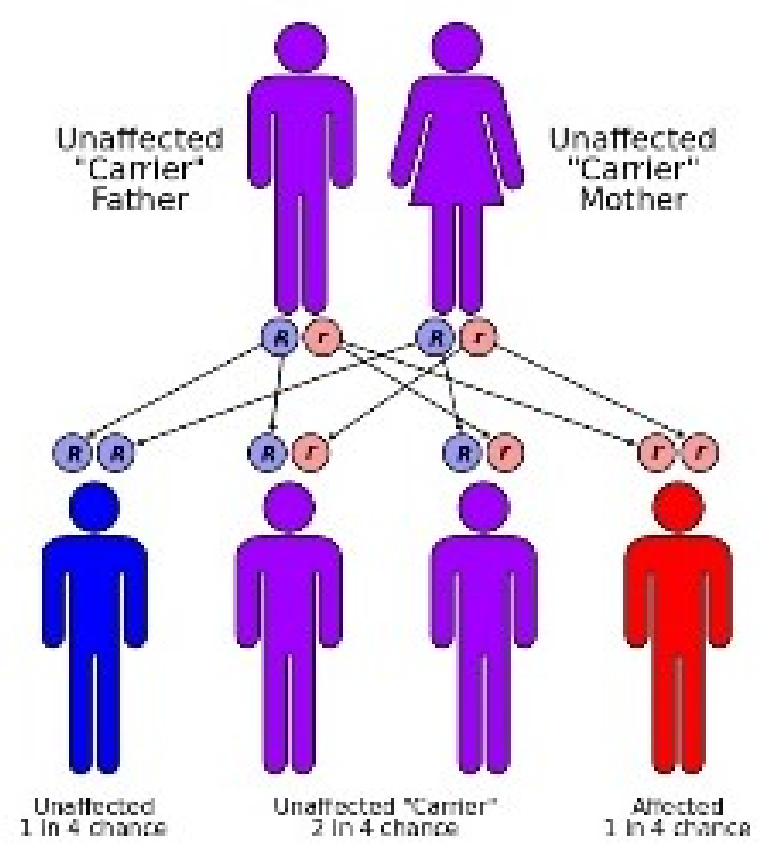
- A person who has one faulty allele will not have the disease, as one allele will still produce the correct protein, but they can pass that allele on to offspring.

Learning Aim: List and briefly discuss examples of inherited disease from gene mutation.

Inherited disease from gene mutation
Example: Albinism

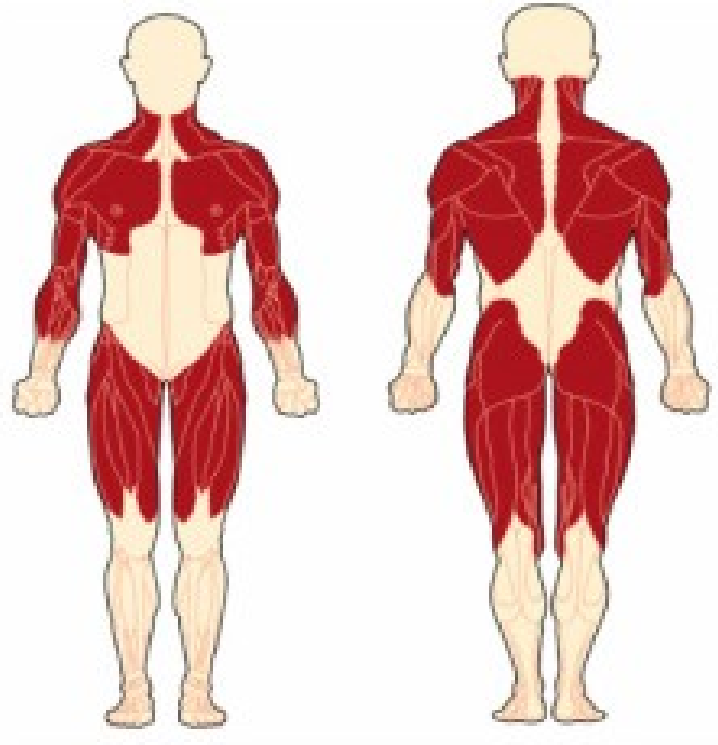
- Allele for pigment production is faulty
- People who inherit two faulty alleles will have *albinism*. (Autosomal recessive inheritance)
  - Cannot produce pigment for skin and hair
    - White blond hair
    - Extremely pale skin
    - Extremely pale blue eyes





Learning Aim: Discuss the genetic cause of albinism and features of albinism

Inherited disease from gene mutation
Example: Duchenne Muscular Dystrophy

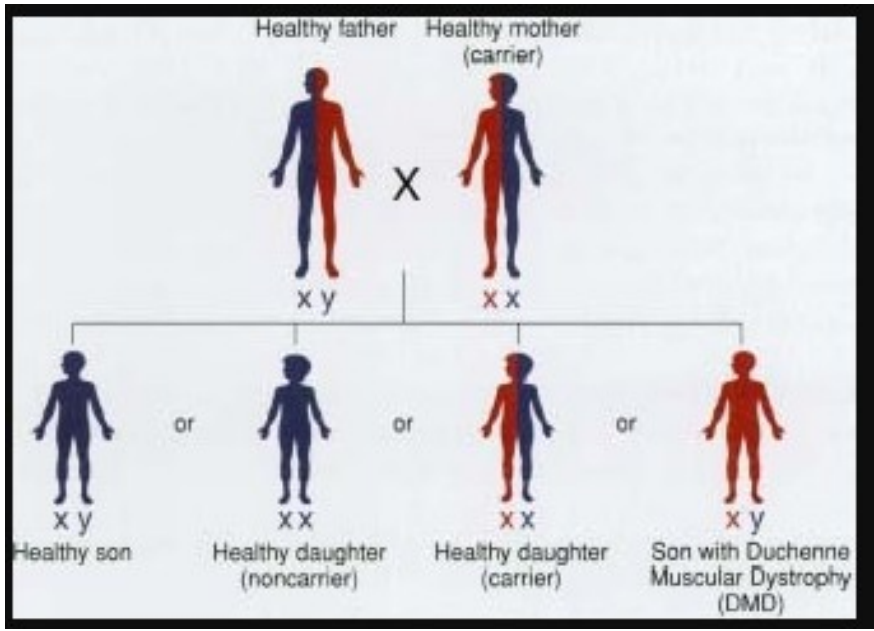


- Faulty gene allele for a protein required for muscle development, on chromosome pair 23. Usually caused by base deletion.
  - Wasting of leg muscles, arm, shoulders and chest
  - Death due to respiratory failure
  - Survival unlikely past 20 years old.

- Recessive – if one healthy allele is present, protein can still be made.

- Mother is carrier, can pass on disease to her sons.
  - X-linked inheritance

- Can also occur as a spontaneous mutation in gamete/zygote.



Learning Aim: Discuss the genetic cause of DMD and symptoms of DMD

Inherited disease from gene mutation
Example: Cystic Fibrosis

- Mutation on chromosome 7.

- Faulty allele for gene (1480 amino acids long) that regulates chloride ion movement across cell membrane. Usually caused by base deletion.

- Autosomal recessive inheritance: if one normal allele is present, protein will still be made.

- Affected individuals:
  - Mucus is stickier than unaffected individuals
  - Causes problems with respiratory and digestive systems

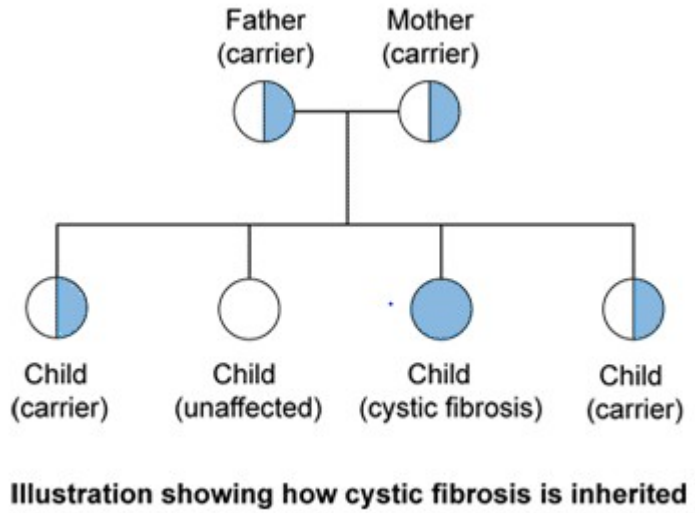
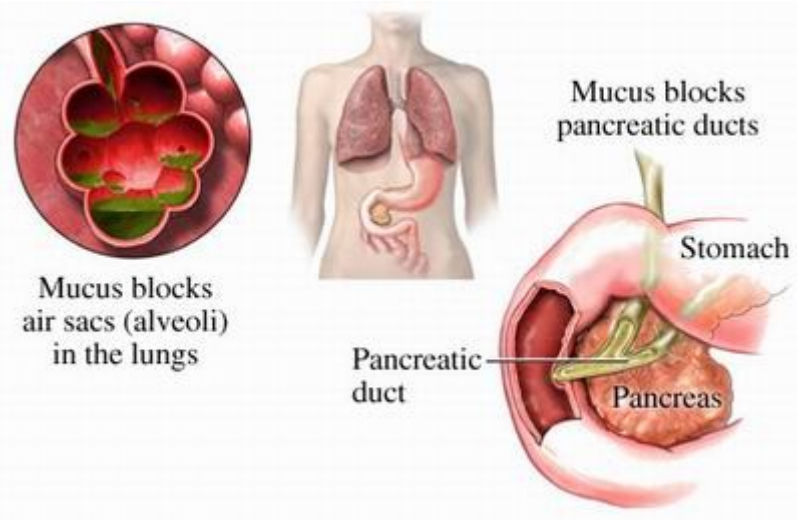


Illustration showing how cystic fibrosis is inherited



Learning Aim: Discuss the genetic cause of CF and symptoms of CF

Inherited disease from gene mutation – Lethal Recessives
Example: Tay-Sachs Disease

- Some autosomal recessive conditions (where the condition occurs if the child inherits two faulty alleles for a gene) are lethal.
  - Cause death of embryo, foetus or very young child

  Eg: Tay-Sachs Disease (TSD)
    - Disorder of lipid metabolism – faulty code for an important enzyme
    - Causes accumulation of fatty substance in brain
    - Death occurs in early childhood
    - Must inherit two faulty alleles to be affected. If only one inherited, person will be a carrier of the faulty allele, but can still make the enzyme.

Learning Aim: Define the term "lethal recessive" and give an example.

Inherited disease from gene mutation – Lethal Recessives
Example: Tay-Sachs Disease

- Because people who are affected die before they can reproduce, proportion of lethal recessive alleles in the gene pool will slowly reduce, unless the population is small (with a higher chance of two carriers reproducing).

- Tay-Sachs mutation exists at a higher frequency in Ashkenazi Jews, a small population who usually marry others from the group

- Gene frequency of the lethal allele is higher in these populations than the general population.
  - 1 in 29 Ashkenazi Jews carry the Tay-Sachs allele
  - 1 in in 250 people in the general population carry the Tay-Sachs allele.

Learning Aim: Define the term "lethal recessive" and give an example.